

Technology #160

A lightweight, low-drag monocoque passenger car with a teardrop-inspired body, full underbody and wheel-arch fairings, active grille shutters, camera-based side mirrors, narrow low-rolling-resistance tires, and a small highly efficient hybrid powertrain with electric accessory drive and engine

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-09

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

A lightweight, low-drag monocoque passenger car with a teardrop-inspired body, full underbody and wheel-arch fairings, active grille shutters, camera-based side mirrors, narrow low-rolling-resistance tires, and a small highly efficient hybrid powertrain with electric accessory drive and engine start-stop plus pulse-and-glide capable control.

WHY DOES IT MATTER?

Documented problem 'aerodynamic drag and powertrain energy loss' concerning passenger car. User-submitted engineering problem (session input): "create the worlds most fuel efficient car". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 23 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 5-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

constraint-compliance envelope article

WHAT WOULD MAKE US KILL IT?

killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 5-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: constraint-compliance envelope article

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - constraint-compliance envelope article - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: constraint-compliance characterization. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-002 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-004 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: failure-mode stress tests (distribution shift). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII.); Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: constraint-compliance envelope article. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Reducing aerodynamic drag force (proportional to $C_d \cdot A \cdot v^2$) and parasitic powertrain losses by streamlining the body, minimizing frontal area, and decoupling accessory loads from the engine.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'aerodynamic drag and powertrain energy loss' concerning passenger car. User-submitted engineering problem (session input): "create the worlds most fuel efficient car". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: nhtsa_recalls: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [nhtsa_recalls] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: A lightweight, low-drag monocoque passenger car with a teardrop-inspired body, full underbody and wheel-arch fairings, active grille shutters, camera-based side mirrors, narrow low-rolling-resistance tires, and a small highly efficient hybrid powertrain with electric accessory drive and engine start-stop plus pulse-and-glide capable control.

Subsystems:

- ?: data ingestion and labeling pipeline - ?
- ?: feature / representation layer - ?
- ?: model + uncertainty quantification - ?
- ?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: A lightweight, low-drag monocoque passenger car with a teardrop-inspired body, full underbody and wheel-arch fairings, active grille shutters, camera-based side mirrors, narrow low-rolling-resistance tires, and a small highly efficient hybrid powertrain with electric accessory drive and engine start-stop plus pulse-and-glide capable control.

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: R_{emp} , loss.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: APPLICABLE; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: U.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'aerodynamic drag and powertrain energy loss', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	must remain a practical road-legal passenger car	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Reducing aerodynamic drag force (proportional to $C_d \cdot A \cdot v^2$) and parasitic power losses by streamlining the body, minimizing the frontal area, and reducing the weight of the vehicle	MODELLER
DI-003	Expected effect target	Highway fuel consumption below 1.5 L/100 km gasoline-equivalent; drag coefficient below 0.15; total road-load energy per unit distance below 100 kJ/km	MODELLER
DI-004	Domain design input pattern: decision tree applicability	US EPA support tokens [‘decision’]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: available applicability	NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: latent variable applicability	NOT ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards	EU Emissions Regulation (EU 2019/1242) and UNECE R101 (Global Technical Regulation for Light Duty Vehicles)	ENGINEERING_PROPOSED
DI-008	Custodied observation: Lightweighting in automotive	https://autonng.com/2022/01/01/industry-2022-reports-on-lightweighting/	MODELLER
DI-009	Custodied observation: Total CO ₂ equivalent life cycle	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-010	Custodied observation: Comprehensive life cycle	https://danone.com/2022/01/01/industry-2022-reports-on-lightweighting/	MODELLER
DI-011	Custodied observation: Aerodynamic sources	https://daichigaku.com/2022/01/01/industry-2022-reports-on-lightweighting/	MODELLER
DI-012	Custodied observation: Vehicle Aerodynamics	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-013	Custodied observation: Passenger Car Aerodynamic Drag	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-014	Custodied observation: Aerodynamic drag analysis	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-015	Custodied observation: Estimating passenger loading	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-016	Custodied observation: Comprehensive Energy Footprint	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-017	Custodied observation: Drive Aerodynamics	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-018	Custodied observation: ALGO Mobility Systems	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-019	Custodied observation: Next-Generation Single-Passenger	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-020	Custodied observation: Aerodynamic drag reduction	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-021	Custodied observation: Investigation of retention	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-022	Custodied observation: Computational analysis of the F107	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER
DI-023	Custodied observation: Open-World Aerodynamics	https://www.energen.com/2022/11/21/energy-availability-2022/	MODELLER

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings	ONCE PROPOSED	data ingestion and labeling pipeline realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings
DO-002	feature / representation layer realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings	ONCE PROPOSED	feature / representation layer realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings
DO-003	model + uncertainty quantification realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings	ONCE PROPOSED	model + uncertainty quantification realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings
DO-004	deployment target + monitoring loop realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings	ONCE PROPOSED	deployment target + monitoring loop realizing A lightweight, low-drag monocoque passenger car with a first drop-inspired body, full underbody and wheel-arch fairings
DO-005	data ingestion and labeling pipeline interfaces with feature / representation layer	ONCE PROPOSED	data ingestion and labeling pipeline interfaces with feature / representation layer
DO-006	feature / representation layer interfaces with model + uncertainty quantification	ONCE PROPOSED	feature / representation layer interfaces with model + uncertainty quantification
DO-007	model + uncertainty quantification interfaces with deployment target + monitoring loop	ONCE PROPOSED	model + uncertainty quantification interfaces with deployment target + monitoring loop
DO-008	Device boundaries: patient/fluid inlet and outlet terminal	ONCE PROPOSED	Device boundaries: patient/fluid inlet and outlet terminal
DO-009	Flow/force/signal path geometry realizing the mechanism	ONCE PROPOSED	Flow/force/signal path geometry realizing the mechanism

ID	Description	Status	Missing Inputs
DO-010	Operating range for training data volume and provenance	UNKNOWN	Which the mechanism holds its stated behavior for training data volume and provenance;
DO-011	Operating range for label quality rate within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for label quality rate; verification of range e
DO-012	Operating range for target operating point (precision/recall)	UNKNOWN	Which the mechanism holds its stated behavior for target operating point (precision/recall);
DO-013	Operating range for calibration error budget within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for calibration error budget; verification of
DO-014	Operating range for inference latency ceiling within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for inference latency ceiling; verification of
DO-015	Operating range for distribution-shift monitoring thresholds	UNKNOWN	Which the mechanism holds its stated behavior for distribution-shift monitoring thresholds;
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating point	PROPOSED	ensuring high data integrity, selection of a decision model that passes (operational) test equipment
DO-017	Information pipeline: ingestion -> feature/representation -> UQ -> decision	PROPOSED	stage input/output timing loop, latency budget, timeslotting, persistence, specification
DO-018	Fixture enabling: held-out + temporal validation splits	UNKNOWN	Concept named, build specification, UNKNOWN selection; acceptance instrumentation
DO-019	Fixture enabling: calibration and decision-threshold analysis	UNKNOWN	Concept named, build specification, UNKNOWN selection; acceptance instrumentation
DO-020	Fixture enabling: failure-mode stress tests (distribution shift)	UNKNOWN	Concept named, build specification, UNKNOWN selection; acceptance instrumentation
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Concept named, build specification, UNKNOWN selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and distribution	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
target operating point (pressure)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method
distribution-shift monitoring thresholds	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as sourced candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty	
aerodynamic drag and powertrain efficiency loss	Failure to accurately model drag and powertrain efficiency	design verification, ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Endo)	
distribution shift degradation by misdiagnosis deployment	Failure by misdiagnosis deployment	designing drag and yet to be deployed	ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Arts)
label noise propagating into base prediction	Failure by label noise propagating into base prediction	designing drag and yet to be deployed	ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Arts)
overfitting to source-side data	Failure by overfitting to source-side data	designing drag and yet to be deployed	ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Arts)
calibration drift invalidating decision thresholds	Failure by calibration drift invalidating decision thresholds	designing drag and yet to be deployed	ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Arts)
feedback-loop data contamination	Failure by feedback-loop data contamination	designing drag and yet to be deployed	ETO, SSS, BVS, HVT, etc. requires design to work	UNKNOWN	Verification, test, degraded data failure mode (Arts)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
aerodynamic drag and powertrain energy loss	Variable aerodynamic drag and powertrain energy loss	designer controls NOE ESTABLISHED	data collection - workshop, triggered data failure reports (AmX & V)
distribution shift degrading accuracy	Failure by distribution shift degrading accuracy	many prototyping constraints trigger deficiency in test data collection	test data integrity, failure reports (AmX & V)
label noise propagating to model predictions	Failure by label noise propagating to model predictions	data provenance system enforces labels (USN & V)	data provenance, failure reports (AmX & V)
overfitting to source-site artifacts	Failure by overfitting to source-site artifacts	multisite real-world data acquisition	data provenance, failure reports (AmX & V)
calibration drift invalidating decision thresholds	Failure by calibration drift invalidating decision thresholds	data provenance, failure reports (AmX & V)	data provenance, failure reports (AmX & V)
feedback-loop data contamination	Failure by feedback-loop data contamination	data provenance, failure reports (AmX & V)	data provenance, failure reports (AmX & V)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Build a full-scale rolling body mockup	Build a full-scale rolling body mockup	pass regulatory certification	OK (PASS) The mockup of the rolling body
VF-002	held-out + temporal validation splits.	Acceptance criterion (ENGINEERING_PROPOSED) pre-registered	OK (PASS) The test results	OK (PASS) The test results
VF-003	held-out + temporal validation splits.	Acceptance criterion (ENGINEERING_PROPOSED) pre-registered	OK (PASS) The test results	OK (PASS) The test results
VF-004	failure-mode stress tests (distribution shift)	Acceptance criterion (ENGINEERING_PROPOSED) pre-registered	OK (PASS) The test results	OK (PASS) The test results

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification passes)	criteria to be fixed at the end of the project	NOT PERFORMED Definition; validation is

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data)	NOT	ESTABLISHED (no domain data)				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	constraint-compliance envelope per article method	constraint-compliance envelope per article method	constraint-compliance envelope per article method	NO TESTS ESTABLISHED (to be quoted)	constraint-compliance envelope per article method	NO TESTS ESTABLISHED (to be quoted)
WP-02	article for VF-001	per method	Build a full-scale rolling contact fatigue test rig	NO TESTS ESTABLISHED (to be quoted)	VF-001 verification test results (to be quoted)	NO TESTS ESTABLISHED (to be quoted)
WP-03	article for VF-002	per method	held-out + temporal variation test	NO TESTS ESTABLISHED (to be quoted)	VF-002 verification test results (to be quoted)	NO TESTS ESTABLISHED (to be quoted)
WP-04	article for VF-003	per method	held-out + temporal variation test	NO TESTS ESTABLISHED (to be quoted)	VF-003 verification test results (to be quoted)	NO TESTS ESTABLISHED (to be quoted)
WP-05	article for VF-004	per method	failure-mode stress test	NO TESTS ESTABLISHED (to be quoted)	VF-004 verification test results (to be quoted)	NO TESTS ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 5-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.